

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++

Tools/Frameworks: TensorFlow, PyTorch, Scikit-learn, Docker, Kubernetes, AWS SageMaker, FastAPI, Apache Spark, Airflow, Git, Linux

Competence: Data Processing Engineering, Machine Learning, MLOps, Deep Learning, Computer Vision, Signal Processing, System Engineering Lifecycle, Earth Observation, Distributed Systems, Cloud Computing, Operational Monitoring, ETL Pipelines, CI/CD

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological data processing applications, optimizing model performance and training efficiency.
- Explored and integrated novel architectures and physics-informed hybrid loss functions, enhancing model accuracy and stability for data processing tasks.
- Collaborated with PhD researchers on metrological data processing for a FLOMEKO conference publication, contributing to research dissemination.
- Developed interactive dashboards for real-time data visualization and model performance monitoring, aiding operational oversight and troubleshooting.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Optimized vitiligo segmentation accuracy to 94% and increased processing speed by 5x using advanced AI algorithms, deployed efficiently with Docker and FastAPI.
- Developed a real-time tennis foul detection system with TensorRT, achieving a 6x reduction in compute cost and processing time, enhancing operational efficiency for sports analytics.
- Managed MLOps pipelines on AWS SageMaker, including monitoring and addressing YOLO model data drift, and maintained Kubernetes infrastructure for robust operational data processing.
- Developed Proof-of-Concepts (POCs) leveraging YOLO, few-shot learning, and CNNs for diverse client applications, demonstrating capabilities in data processing and system integration.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack web application providing AI-driven data processing services, using Flask to manage backend operations and data workflows.

- Implemented an AI chatbot application leveraging the OpenAI API, focusing on efficient text data processing and integration within a system environment.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented robust web scrapers and ETL pipelines for efficient processing of structured and unstructured data, ensuring data quality for analytics and operational use cases.
- Automated data ingestion and cleaning workflows, enhancing data reliability and reducing manual effort for client analytics and operational use cases.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble (DPFunc, DeepGOPlus) for protein function prediction, utilizing AlphaFold for feature extraction and mitigating taxonomic bias in biological data processing.
- Enhanced model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces, demonstrating advanced data manipulation skills and V&V principles.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a real-time forecasting system for Bitcoin prices, integrating various data sources and applying advanced time-series analysis.
- Implemented data pipelines for continuous ingestion and processing of market data, ensuring timely and accurate predictions for operational insights.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Collaborated on the world's first AI-driven self-playing Chitralli Sitar, integrating advanced signal processing and music generation LLMs for complex data synthesis.
- Achieved significant visibility, with the project featured on Coke Studio, showcasing innovative application of AI and creative engineering.

Hugging Face

Open-Source

Contributor

- Co-authored technical documentation and implementation for Zero-shot Image Classification on Hugging Face Hub, contributing to knowledge sharing and standardization within the open-source community.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Relevant Coursework: Advanced Data Processing, Machine Learning Systems, Distributed AI

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Thesis Project: Developed Real-time Bitcoin Price Forecasting system (Razanalytics)

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python